

## Unit 203: Engineering mathematics and science principles

### Handout 1: BODMAS and solving simple equations

#### Transposition of formulae and solving simple equations

Engineering can be divided into a wide range of specialist areas of work such as mechanical, electrical/electronic, chemical and civil. One common factor to all areas of engineering is the use of formulae to calculate values of required quantities or properties. A formula is an algebraic rule. It always has an equals sign. The skill of manipulating a formula is known as **transposition**. There are some simple rules to follow to transpose formulae reliably and correctly.

A method known as BODMAS is used to make sure that a formula is rearranged (transposed) in the right order to avoid changing its meaning.

Work out **Brackets** first, then **Of** such as **squaring of** and **square roots of**, then **Divide/Multiply** groups of numbers before **Adding** or **Subtracting** them. This set of rules works really well for simple cases, so remember the word: BODMAS.

#### Examples

1. Solve the equation  $X = 21 + 4 - (6 - 3)^2$

Step 1: Brackets  $X = 21 + 4 - 3^2$

Step 2: Order (square)  $X = 21 + 4 - 9$

Step 3: Addition/subtraction  $X = 25 - 9 = 16$

2.  $Q = (R - 5)^2 + \frac{7T}{W}$ . Find the value of  $Q$  when  $R = 8$ ,  $T = 3$  and  $W = -7$ .

Step 1: Brackets  $Q = (8 - 5)^2 + 7 \times \frac{3}{-7}$

Step 2: Squared and multiple  $Q = 3^2 + \frac{21}{-7}$

Step 3: Adding/subtracting  $Q = 9 + -3$

$$Q = 9 - 3 = 6$$

3. Rearrange the formula  $A = \frac{\pi d^2}{4}$  to make  $d$  the subject.

Step 1: Multiple both sides by 4  $4A = \pi d^2$

Step 2: Divide both sides by  $\pi$   $\frac{4A}{\pi} = d^2$

Step 3: Square root both sides  $\sqrt{\frac{4A}{\pi}} = d$

$$d = \sqrt{\frac{4A}{\pi}}$$

4. Rearrange the formulae  $c(a - 4) = b(4 - 2a)$  to make  $a$  the subject.

Step 1: Work out the brackets

$$ac - 4c = 4b - 2ab$$

Step 2: Group like terms together and simplify

$$ac + 2ab = 4b + 4c$$

$$a(c + 2b) = 4(b + c)$$

Step 3: Divide both sides by  $(c + 2b)$

$$a = 4(b + c) / (c + 2b)$$

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### Handout 2: Trigonometry

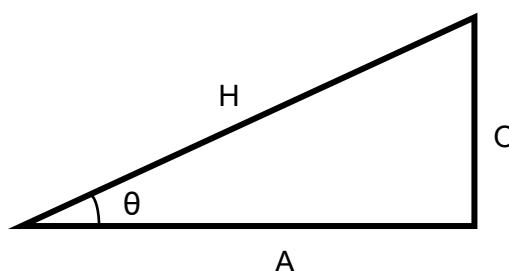
It is often necessary to calculate the angles or lengths of sides in a triangle. For example, this may be needed to mark out a part, to determine the path for a machine tool, or to calculate the angle of a taper on a turned part.

For a right-angled triangle, the angle  $\theta$  is related to the hypotenuse (H), opposite (O), and adjacent (A) by the equations for the tangent (tan), sine (sin) and cosine (cos).

$$\tan \theta = O \div A$$

$$\sin \theta = O \div H$$

$$\cos \theta = A \div H$$



If at least two values are known, these equations can be rearranged to find the unknown values:

$$\theta = \tan^{-1} (O \div A)$$

$$A = O \div (\tan \theta)$$

or

$$A = H \times (\cos \theta)$$

$$\theta = \sin^{-1} (O \div H)$$

$$O = A \times (\tan \theta)$$

or

$$O = H \times (\sin \theta)$$

$$\theta = \cos^{-1} (A \div H)$$

$$H = A \div (\cos \theta)$$

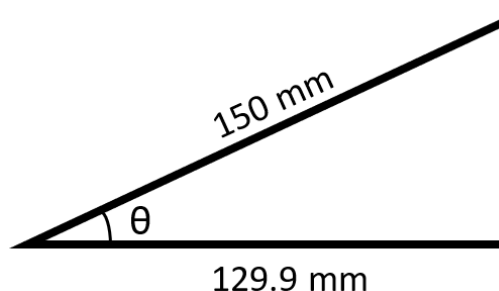
or

$$H = O \div (\sin \theta)$$

#### Example

The right-angled triangle shown needs to be marked out on a sheet of material.

Calculate the angle  $\theta$ .



#### Answer

$$\theta = \cos^{-1} (A \div H),$$

$$\theta = \cos^{-1} (129.9 / 150) = \cos^{-1} (0.866)$$

$$\theta = 30^\circ$$

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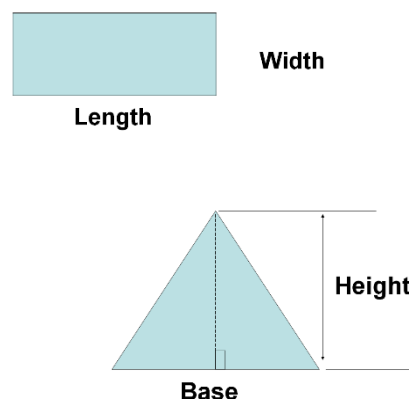
### Handout 3: Shapes

The area, perimeter or volume of a component may be needed to calculate, for example:

- the size or weight of material needed to make a number of parts
- how much waste will be created when a part is cut out
- how much material is held in a container.

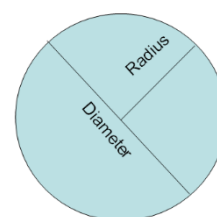
The formulae for the areas of simple shapes are:

- Area of a rectangle = length  $\times$  width =  $L \times W$
- Area of a square = width  $\times$  width =  $W^2$
- Area of a triangle = half (base  $\times$  height) =  $\frac{1}{2} (B \times H)$
- Area of a circle =  $\pi r^2$



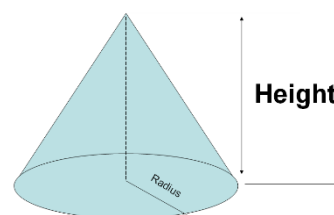
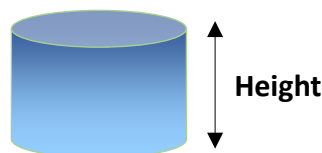
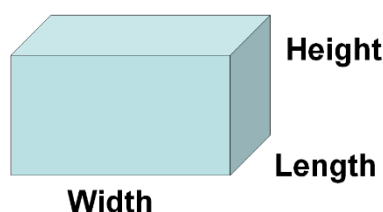
The formulae for the perimeters of simple shapes are:

- Perimeter of a rectangle =  $(2 \times \text{length}) + (2 \times \text{width})$
- Perimeter of a square =  $4 \times \text{width}$
- Perimeter of a triangle = side 1 + side 2 + side 3)
- Circumference (perimeter) of a circle =  $2 \pi r$



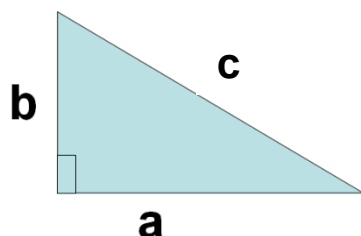
The formulae for the volumes of simple shapes are:

- Volume of a cuboid,  $V = \text{length} \times \text{width} \times \text{height} = L \times W \times H$
- Volume of a cylinder,  $V = \text{area of circle} \times \text{height} = A \times h = \pi r^2 \times h$
- Volume of a sphere,  $V = \frac{4}{3} \pi r^2$
- Volume of a cone,  $V = \frac{1}{3} \times \text{area of circle} \times \text{height} = \frac{1}{3} \pi r^2 \times H$



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### Handout 4: Pythagoras' theorem



$$a^2 + b^2 = c^2$$

$$3 \times 3 = 9$$

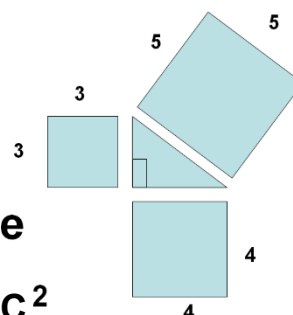
$$4 \times 4 = 16$$

$$5 \times 5 = 25$$

#### Example

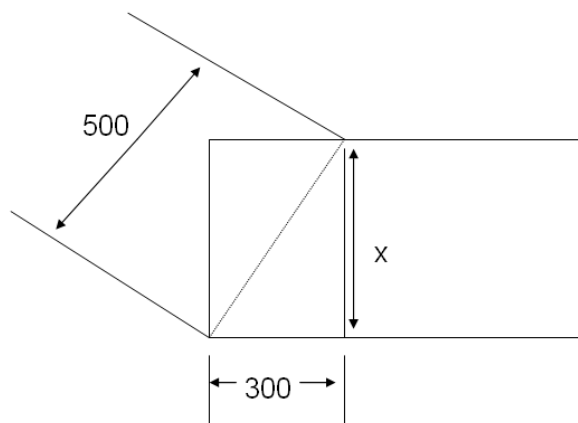
$$a^2 + b^2 = c^2$$

$$9 + 16 = 25$$



#### Example

In the diagram shown opposite, two metal plates are going to be spot welded together. What is the value of the length  $x$  (the width of the metal plate)?



Using  $a^2 + b^2 = c^2$

$$300^2 + x^2 = 500^2$$

$$90\,000 + x^2 = 250\,000$$

$$x^2 = 250\,000 - 90\,000 = 160\,000$$

$$x = \sqrt{160\,000} = 400$$

#### Other uses of 3:4:5 triangles

The right-angled triangle with sides that are three, four and five units long is very useful. It can be used to set out a right angle accurately by measuring the length of the three sides.

For example, a piece of rope that is 12 metres long can have two markers or knots tied at three metres and seven metres from one of its ends. The rope is then laid out on the ground in the shape of a right-angle triangle with the two ends of the rope touching and the markers positioned at the two other corners. This technique can be used by groundskeepers to set out the corners of football and rugby pitches. Similarly, if a partition wall needs to set out at right-angles to an existing wall this 'rope trick' can be used.

## Unit 203: Engineering mathematics and science principles

### Handout 5: Terms and units

| Term         | Meaning                                                                         | Units                                                                              |
|--------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Length       | The measurement of something from end to end                                    | metre (m)                                                                          |
| Time         | The continued progress of existence and events in the past, present, and future | second (s)                                                                         |
| Mass         | The amount of matter in an object                                               | kilogramme (kg)                                                                    |
| Density      | The mass of an object divided by its volume                                     | mass per unit volume ( $\text{kg/m}^3$ or $\text{kg m}^{-3}$ )                     |
| Force        | The push or pull on an object that causes it to change velocity                 | newton (N)                                                                         |
| Moment       | The turning effect of a force                                                   | newton metre (N m)                                                                 |
| Pressure     | The force per unit area                                                         | pascal (Pa) or<br>newton per meter squared ( $\text{N/m}^2$ or $\text{N m}^{-2}$ ) |
| Velocity     | The rate at which the position of an object changes                             | metre per second ( $\text{m/s}$ or $\text{m s}^{-1}$ )                             |
| Acceleration | The increase in speed                                                           | metre per second squared ( $\text{m/s}^2$ or $\text{m s}^{-2}$ )                   |
| Efficiency   | The ratio of the useful output to the total input in any system                 | no units - it is a ratio or percentage                                             |

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### Handout 6: Calculations

| Term            | How to calculate                                        | Formula                                                     | Units                                       |
|-----------------|---------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------|
| Mass, $m$       | density $\times$ volume                                 | $m = \rho V$                                                | kg                                          |
| Density, $\rho$ | mass $\div$ volume                                      | $\rho = \frac{m}{V}$                                        | Kg/m <sup>3</sup> or kg m <sup>-3</sup>     |
| Force           | Mass $\times$ acceleration                              | $F = m a$                                                   | N                                           |
| Moment          | Force $\times$ distance from where the force is applied | $Moment = F r$                                              | N m                                         |
| Pressure        | force $\div$ area                                       | $P = \frac{F}{a}$                                           | Pa or N/m <sup>2</sup> or N m <sup>-2</sup> |
| Velocity        | change in position (or distance) $\div$ time            | $v = \frac{d}{t}$                                           | m/s or m s <sup>-1</sup>                    |
| Acceleration    | change in velocity $\div$ time                          | $a = \frac{v_2 - v_1}{t}$                                   | m/s <sup>2</sup> or m s <sup>-2</sup>       |
| Efficiency      | output $\div$ input                                     | $Efficiency = (\text{output} \div \text{input}) \times 100$ | no units                                    |

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### Handout 7: Density

Density can be used to calculate:

- the amount of material required to make a number of parts
- the cost of the material in a part.

It is defined as the amount of mass in a particular volume. Mass is defined as the amount of matter in an object:

$$\text{density} = \text{mass} \div \text{volume} = \text{kg/m}^3$$

For example, liquid water has a density of  $1000 \text{ kg/m}^3$ . When it freezes, its density changes to  $920 \text{ kg/m}^3$ . This means that a given mass of water will have more volume, which is why water pipes burst when they freeze in the winter.)

The density of steel is  $7850 \text{ kg m}^{-3}$ . The density of stainless steel is  $8300 \text{ kg m}^{-3}$ .

#### Example

A metal frame manufactured out of stainless steel has a volume of  $0.5 \text{ m}^3$ . What would be its approximate mass in kilograms?

$$\text{density} = \text{mass} \div \text{volume}$$

$$\text{Rearranging the formula: mass} = \text{density} \times \text{volume}$$

$$\text{Mass} = 8300 \times 0.5 = 4150 \text{ kg}$$

#### Relative density

Relative density is a way of comparing the density of a material in comparison with that of water.

Relative density has no units because it is a ratio. A value greater than one means that the material is denser than water. A value less than one means that the material is less dense than water.

The relative density of steel is  $7850 \div 1000 = 7.85$ . The relative density of stainless steel is  $8300 \div 1000 = 8.3$ .

| Material | RD   | Material | RD    | Material   | RD  | Material | RD   |
|----------|------|----------|-------|------------|-----|----------|------|
| Zinc     | 7.05 | Mercury  | 13.61 | Aluminium  | 2.7 | Brass    | 8.56 |
| Gold     | 19.3 | Lead     | 11.35 | Balsa wood | 0.2 | Chromium | 6.86 |



## Unit 203: Engineering mathematics and science principles

### Handout 8: Specific heat

Heat and temperature are not the same thing, but they are related to each other.

- Heat is the **total energy** in a substance due to molecular motion.
- Temperature is a measure of the **average energy** of individual molecules in a substance.

#### Specific heat

Specific heat is heat that causes a change in the temperature of an object. The amount of heat energy,  $Q$ , required to raise the temperature of a substance depends upon:

- the mass,  $m$ , of the substance — the larger the mass, the greater the heat energy required
- the specific heat capacity,  $c$ , of the substance.
- the temperature change,  $\Delta t$  — the greater the change in temperature, the greater the heat energy required.

This is represented by the formula:

$$\text{Quantity of heat energy} = \text{mass} \times \text{specific heat capacity} \times \text{temperature change}$$

$$\text{or } Q = mc\Delta t$$

The SI unit for quantity of heat is the joule (J). The units kilo joule ( $1 \text{ kJ} = 10^3 \text{ J}$ ) and mega joule ( $1 \text{ MJ} = 10^6 \text{ J}$ ) are also used.

#### Specific heat capacity

The specific heat capacity is the amount of heat required to change a unit mass of a substance by one unit of temperature. For example, one kilogram of water requires 4187 joules of heat energy to raise its temperature by one degree Celsius. Thus, the specific heat capacity of water is 4187 joules per kilogram per one degree Celsius. This value is normally stated as  $4187 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ .

Typical values of the specific heat capacity,  $c$ :

| Substance | Specific heat capacity, $\text{J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ |
|-----------|------------------------------------------------------------------------|
| Water     | 4187                                                                   |
| Cast iron | 544                                                                    |
| Steel     | 480                                                                    |
| Copper    | 385                                                                    |
| Aluminium | 921                                                                    |

#### Examples

Calculate the amount of heat, in mega joules, that is necessary to raise 5 kg of water from  $20^\circ\text{C}$  to  $100^\circ\text{C}$ . Take the specific heat capacity of water to be  $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ .

$$Q = mc\Delta t = 5 \times 4200 \times (100 - 20) = 1680000 \text{ J} = 1.68 \text{ MJ}$$

Calculate the amount of heat, in mega joules, that is necessary to raise 12 kg of steel from  $30^\circ\text{C}$  to  $730^\circ\text{C}$ . Take the specific heat capacity steel to be  $480 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$ .

$$Q = mc\Delta t = 12 \times 480 \times (730 - 30) = 4032000 \text{ J} = 4.032 \text{ MJ}$$

## Unit 203: Engineering mathematics and science principles

### Handout 9: Latent heat

#### Latent heat

Latent heat is the amount of energy needed to change the state of a material; for example, from solid to liquid or from liquid to gas. When a change of state is from solid to liquid this is called the **specific latent heat of fusion**. When a change of state is from **liquid to vapour** this is called the **specific latent heat of vaporisation**.

For example, 1 kg of water at a temperature of 100°C requires 2257 kJ of heat energy to convert it into steam at the same temperature, i.e. the specific latent heat of vaporisation of water is 2257 kJ kg<sup>-1</sup>.

As energy can neither be created nor destroyed, upon cooling the same amount of heat energy is given up by the substance during transformation.

The table gives typical values of latent heat,  $h$ , at atmospheric pressure:

| Substance | Specific latent heat of fusion, kJ kg <sup>-1</sup> | Specific latent heat of vaporisation, kJ kg <sup>-1</sup> |
|-----------|-----------------------------------------------------|-----------------------------------------------------------|
| Water     | 334                                                 | 2257                                                      |
| Oxygen    | -                                                   | 243                                                       |
| Aluminium | 9130                                                |                                                           |
| Copper    | 3850                                                |                                                           |
| Steel     | 4200                                                |                                                           |

The quantity of latent heat,  $Q$  required to change the state of a substance depends upon:

- the mass,  $m$ , of the substance—the larger the mass, the greater the heat energy required
- the specific latent heat,  $h$ , of the substance.

This is represented by the formula:

**Quantity of latent heat energy = mass × latent heat**

or  $Q = m h$

#### Example

The specific latent heat required to melt ice is 335 kJ kg<sup>-1</sup>.

Find the quantity of latent heat required to melt 3 kg of ice at 0°C.

$$Q = m h$$

$$Q = 3 \times 335000$$

$$Q = 1005000 \text{ J} = 1.005 \text{ MJ}$$

## Unit 203: Engineering mathematics and science principles

### Handout 10: Coefficient of expansion

#### Expansion and contraction due to change in temperature

Most substances increase in volume when heated. This expansion has many uses in engineering.

For example, thermal expansion is used in mechanical applications to fit parts over one another. A bushing can be fitted over a shaft by making its inner diameter slightly smaller than the diameter of the shaft, then heating it until it fits over the shaft, and allowing it to cool after it has been pushed over the shaft, thus achieving a 'shrink fit'.

The table shows some typical values for *coefficient of linear expansion*. Consider steel for example. For every 1°C that steel is heated it will expand by 0.000011 m for every metre of length. Likewise, it will expand 0.000011 mm for every millimetre of length. If cooled it will also contract by the same amount. 0.000011 can be written as  $11 \times 10^{-6}$ .

| Material  | Coefficient of linear expansion, °C <sup>-1</sup> |
|-----------|---------------------------------------------------|
| Steel     | $11 \times 10^{-6}$                               |
| Copper    | $17 \times 10^{-6}$                               |
| Aluminium | $23 \times 10^{-6}$                               |
| Brass     | $20 \times 10^{-6}$                               |

The symbol for **coefficient of linear expansion** is  $\alpha$  (alpha). A coefficient is a constant and therefore has no units. The symbol for **length** is  $L$ . The symbol for **temperature change** is  $\Delta t$ . The formula for determining the increase (decrease) in length is:

$$\text{Increase (decrease) in length} = \alpha \times L \times \Delta t$$

#### Example

A steel strip is 1.5348 m long at 12°C and is pre-heated to 100°C prior to welding.

How long is the strip of steel at 100°C?

Answer:

$$\text{Increase in length} = \alpha \times L \times \Delta t$$

$$\text{Increase in length} = 11 \times 10^{-6} \times 1.5348 \times 88$$

$$\text{Increase in length} = 0.0014857 \text{ m}$$

$$\text{Length of strip at } 100^\circ\text{C} = \text{original length} + \text{increase in length}$$

$$\text{Length of strip at } 100^\circ\text{C} = 1.5348 \text{ m} + 0.0014857 \text{ m}$$

$$\text{Length of strip at } 100^\circ\text{C} = 1.5363 \text{ m}$$

An increase in length of 1.5 mm.

## Unit 203: Engineering mathematics and science principles

### Handout 11: Material types

#### Metals commonly used in engineering

| Material grouping  | Group description                                                                                 | Group members                      | Member description                                                                                                                                              |
|--------------------|---------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ferrous metals     | Ferrous metals and alloys contain the element iron.                                               | Low, medium and high carbon steels | Versatile and widely used. Higher carbon steels are harder and can be used as cutting tools.                                                                    |
|                    | Steels are alloys of iron and small amounts of other elements, particularly carbon and manganese. | Stainless steel                    | Contains high proportion of chromium to provide corrosion resistance. Used in catering to maintain hygiene standards.                                           |
|                    |                                                                                                   | Cast iron                          | Contains iron and carbon. Does not need high cost refinement. Can be easily cast into complex shapes.                                                           |
| Non-ferrous metals | Do not contain iron.                                                                              | Aluminium and aluminium alloys     | Lightest of commonly used metals. Very good thermal and electrical conductivity. Good corrosion resistance. Mechanically weak and difficult to weld and solder. |
|                    | Each metal has its own properties and uses.                                                       |                                    |                                                                                                                                                                 |
|                    | In general, with the exception of copper, non-ferrous metals are mixed to form alloys.            | Copper                             | Very good thermal and electrical conductivity. Good corrosion resistance. Easy to solder and braze. Relatively strong. Used in electrical cables.               |
|                    | Many are corrosion resistant.                                                                     | Brass                              | Alloy of copper (60%) and zinc (40%). Corrosion resistant. Used for plumbing fittings.                                                                          |
|                    |                                                                                                   | Bronze                             | Alloy of copper (97%), zinc (2.5%) and tin (0.5%). Used for coins.                                                                                              |

### Non-metallic materials commonly used in engineering

| Material grouping | Group description                                                                                                             | Group members                                             | Member description                                                                                                                                                                                                                  |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plastics          | Lightweight, strong relative to weight. Good electrical insulators. Good resistance to corrosion especially acids.            | thermoplastics                                            | Set/reset by heating.<br>Types include acrylic, polycarbonate, polyethylene, ABS, PVC.<br>Used for bowls, children's toys, water bottles.                                                                                           |
|                   |                                                                                                                               | thermosetting                                             | Set by chemical reaction; cannot be reset. Rigid and stronger than thermoplastics.<br>Types include epoxy resin, melamine formaldehyde, urea formaldehyde.<br>Used for plug sockets, coverings for kitchen worktops and pan handles |
| Composites        | Combination of two or more components (typically reinforcement and matrix) to gain advantages of each component's properties. | Glass reinforced polymer (GRP, also known as fibreglass). | Glass fibres in an epoxy or polyester matrix.<br>Combined material is very strong and lightweight.<br>Corrosion resistant. Used for aircraft and car bodies.                                                                        |
|                   |                                                                                                                               | Carbon reinforced polymer (CRP)                           | Carbon fibres in an epoxy or polyester matrix.<br>Tough lightweight material.<br>Used in aerospace, bicycles and motorbikes, sailboats, fishing rods, sports rackets, crash helmets.                                                |
|                   |                                                                                                                               | Aramid fibre reinforced composite                         | Aramid fibres in a polymer matrix.<br>Heat resistant and strong synthetic fibres, very difficult to cut.<br>Used in aerospace and military applications such as body armour, flame-resistant clothing and helmets.                  |
| Natural materials | Elastomer                                                                                                                     | Rubber                                                    | A natural product known for its flexibility and durability. Used for belts, tyres, anti-vibration mountings and drive couplings.                                                                                                    |

## Unit 203: Engineering mathematics and science principles

### Handout 12: Properties of materials

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#### Mechanical properties

| Property            | Description                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------|
| <b>Strength</b>     | The ability of a material to withstand an applied load without failure or plastic deformation.      |
| <b>Toughness</b>    | The ability of a material to withstand an impact without fracturing.                                |
| <b>Hardness</b>     | The ability of a material to resist scratches and wear.                                             |
| <b>Elasticity</b>   | The ability of a material to return to its original shape when the forces acting on it are removed. |
| <b>Ductility</b>    | The ability of a material to be stretched by a tensile force without breaking.                      |
| <b>Malleability</b> | The ability of a material to be bent or shaped by a compressive force without breaking.             |

#### Physical properties

| Property                    | Description                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Corrosion resistance</b> | The ability of a material to withstand damage caused by oxidization or chemical attack.                                                          |
| <b>Conductivity</b>         | The ease at which an electric charge or heat can pass through a material.                                                                        |
| <b>Insulation</b>           | The resistance to the passage of electricity, heat, or sound.                                                                                    |
| <b>Melting point</b>        | The temperature at which a solid will melt and become a liquid.                                                                                  |
| <b>Magnetism</b>            | A force between two metal items where they will attract or repel each other. This only affects certain alloys containing iron, nickel or cobalt. |
| <b>Density</b>              | Mass per unit volume.                                                                                                                            |
| <b>Colour</b>               | A visual quality relating to the light reflected by the object.                                                                                  |